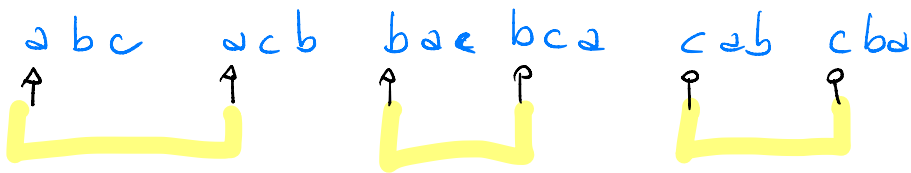


MIN-HASH = min-wise independent permutations

► PERMUTATIONS: a, b, c

X = set of items $|X|!$ perms



Looking at any fixed position, let's say the first one, any item in X appears the same number of times.

If we look at permutations as orderings on X , the first item in the permutation is called minimum

(e.g. $bac \Rightarrow b < a < c \Rightarrow$ the min is b)

$$\text{In our example } pr = \frac{1}{3} = \frac{2}{3!} = \frac{2!}{3!} = \frac{(|X|-1)!}{|X|!} = \frac{1}{|X|}$$

► Let $\mathcal{H}$ be a family of hash functions that "permute" universe U thus assigning an ordering to the elements in U

[in this way we get a ranking h s.t. $a, b \in U : a < b \Leftrightarrow h(a) < h(b)$ for any $h \in \mathcal{H}$]

► Given a set $X \subseteq U$, define $\min h(X) \stackrel{\text{def}}{=} \min_{a \in X} h(a)$

$\mathcal{H}$ is min-wise independent if $\forall X \subseteq U \quad \forall a \in X$:

$$\Pr_{h \in \mathcal{H}} [h(a) = \min h(X)] = \frac{1}{|X|}$$

h chosen uniformly at random

That is, each element $a \in X$ has the same chance to be the min, after X is permuted by h

This is useful to check resemblance among documents, web pages, etc..

document = "bag of words" = we just consider the set of its distinct words

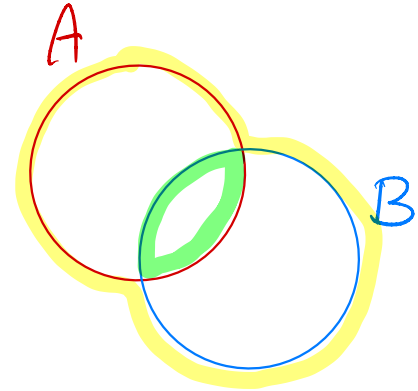
Hence $U = \{ \text{the set of all possible words} \}$
lemmatization (singular, infinite verbal form..)

$D \subseteq U$ document

$A, B \subseteq U$ sets :

► Jaccard's index $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$

$0 \leq J(A, B) \leq 1$
 $\uparrow \qquad \qquad \uparrow$
 $A, B \text{ disjoint} \qquad A=B \Rightarrow A \cap B = A \cup B$



USEFUL in massive collections: $D_1, D_2, \dots, D_d \subseteq U$

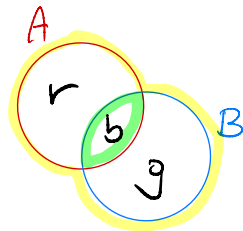
ISSUE $J(D_i, D_j)$ takes $O(|D_i| + |D_j|)$ time for any pair $1 \leq i < j \leq d$

MIN-HASH helps! (Sometimes is called SIM-HASH)

Interesting property:

$$\Pr_{h \in H} (\min h(A) = \min h(B)) = J(A, B) \quad \text{for any } A, B \subseteq U$$

Why this is true?



$$|A| = r + b \quad |B| = g + b$$

$$|A \cap B| = b \quad |A \cup B| = r + b + g$$

Since h is min-wise independent, each element $a \in X = A \cup B$ has the same chance to be the minimum in X under $h \in H$: $\Pr(\min(X) = h(a)) = \frac{1}{|X|} = \frac{1}{r+b+g}$

Now consider such $\min h(a)$: what is the probability that a belongs also to $A \cap B$?

$$\text{It's conditioned probability: } \frac{b}{|X|} = \frac{b}{r+b+g} = \frac{|A \cap B|}{|A \cup B|} = J(A, B)$$

In order to exploit this property, we face two issues:

- ① In many applications, randomly permuting U is unfeasible as U is huge
- ② Using a single $h \in H$ does not give enough precision to approximate J

We have familiar tools to circumvent these issues:

- ① It has been shown that using a universal hashing family for H is "almost" as good as using a random permutation (no proof given here)
- ② As we did for Morris' counters, repeat the experiment k times
(also known as k-min sketch, quite similar)
INDICATOR VARIABLE $h_1, h_2, \dots, h_k \in H$ (universal hash family in practice)
Given A, B : $X_i = \begin{cases} 1 & \text{if } \min h_i(A) = \min h_i(B) \\ 0 & \text{o.w.} \end{cases}$ with $pr = J(A, B)$
 $E[X_i] = J(A, B)$ UNBIASED ESTIMATOR

Let $Y = \frac{\sum_{i=1}^k X_i}{k}$ is still an unbiased estimator ($E[Y] = \frac{1}{k} \sum_{i=1}^k E[X_i] = \frac{k}{k} J(A,B) = \mu$)
 ← needed to get $0 \leq Y \leq 1$ as $J(A,B)$

We use CB to get a FPTAS (ϵ, δ)

FIRST ATTEMPT:

$$Y' = \sum_{i=1}^k X_i \quad (\text{hence } Y' = kY) \Rightarrow \mu' = k\mu \quad (\mu' = k J(A,B))$$

$$CB \Rightarrow \Pr[|Y - \mu| \geq \epsilon \mu] = \Pr[|Y' - \mu'| \geq \epsilon \mu'] \stackrel{CB}{\leq} 2 \cdot e^{-\frac{\epsilon^2 \mu'}{3}} = 2 e^{-\frac{\epsilon^2 k \mu}{3}} \leq \delta$$

$$\text{This gives } k = O(\epsilon^{-2} \cdot \underbrace{\delta^{-1} \mu^{-1}})$$

we don't like it here

as $\mu^{-1} = 1/J(A,B)$ can be very large

SECOND AND BEST ATTEMPT:

We use a variation of CB called AZUMA-HOEFFDING (AH) bounds:

$X_1, X_2, \dots, X_k$ i.i.d. random variables s.t. $\mu = E\left[\frac{\sum_{i=1}^k X_i}{k}\right]$
 $a_i \leq X_i \leq b_i$, let $Y = \frac{\sum_{i=1}^k X_i}{k}$ ↑ our Y

$$\forall \varepsilon: \Pr[|Y - \mu| \geq \varepsilon] \leq 2 e^{-\frac{2k^2\varepsilon^2}{\sum_{i=1}^k (b_i - a_i)^2}}$$

note that no μ here!

IN OUR CASE: X_i indicator $\Rightarrow a_i = 0, b_i = 1$

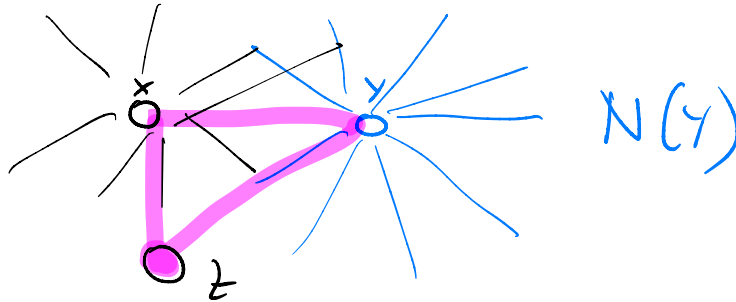
$$\Rightarrow \sum_{i=1}^k (b_i - a_i)^2 = k \Rightarrow 2 e^{-\frac{2k^2\varepsilon^2}{\sum_{i=1}^k (b_i - a_i)^2}} = 2 e^{-\frac{2k^2\varepsilon^2}{k}} = 2 e^{-2k\varepsilon^2} < \delta$$

This gives $k = O(\varepsilon^{-2} \log \delta^{-1})$ GOOD!

APPLICATION: Network analysis

A network is considered "social" if it contains many triangles

Rationale



$$\begin{aligned} \# \text{TRIANGLES:} \\ \sum_{\substack{xy \in E \\ x < y}} |N(x) \cap N(y)| \end{aligned}$$

if x and y are friends, they have also common friends z
and thus they form a triangle $\{x, y, z\}$

This idea is at the heart of some community detection algorithms

Network is a graph $G = (V, E)$

$N(x) = \{x' : xx' \in E\}$ (same $N(y)$) $\Rightarrow$

z common friend iff
 $z \in N(x) \cap N(y)$
i.e. $\{x, y, z\}$ TRIANGLE

How can we estimate $|\underbrace{N(x)}_A \cap \underbrace{N(y)}_B|$?

We use Jaccard $J(A, B)$ and its approximation just seen before, which we denote by $\tilde{J}(A, B)$ (for any given ϵ, δ)

We now formulate $|N(x) \cap N(y)| = |A \cap B|$ in terms of $J(A, B), \tilde{J}(A, B)$

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|} \Leftrightarrow J(A, B) \cdot (|A| + |B| - |A \cap B|) = |A \cap B|$$

$$\Leftrightarrow J(A, B) (|A| + |B|) = (1 + J(A, B)) \cdot |A \cap B| \Leftrightarrow |A \cap B| = \frac{J(A, B) (|A| + |B|)}{1 + J(A, B)}$$

$$\Rightarrow \text{an approximation of } |A \cap B| = \frac{\tilde{J}(A, B) (|A| + |B|)}{1 + \tilde{J}(A, B)}$$

obj note that error ϵ is both at numerator and denominator

□